

Homework 5, 2026

This homework needs to be done on individual basis (that is, it is not a group homework problem).

Please note: in this assignment statement the notation of MLS is used a few times (not necessarily always) without explanation where it is used.

Consider a four degree of freedom robotic manipulator schematically shown in figure 1, in a certain configuration, which for the purposes of this assignment can be considered as its reference configuration. The manipulator has three links. The base, referred to as link 0, is fixed and aligned with the z axis of the base frame S . Link 1 is connected to link 0 through a revolute joint at B . Link 2 is connected with a revolute joint at D to one end of a prismatic joint, the other end of which is connected to the tip of link 1 at C . The tool, with tool frame T , is connected to the tip of link 2 through a third revolute joint. In the reference configuration all three links lie in the y - z plane of frame S , with links 1 and 2 aligned with one another and parallel with the y axis of frame S . Also, in the reference configuration the axes of frame T are parallel to the corresponding axes of frame S , both frames right handed axis systems.

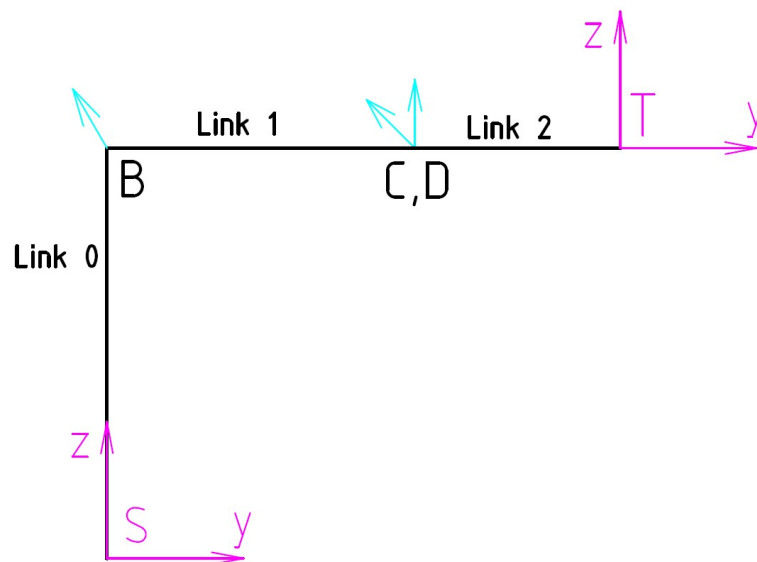


Figure 1: The manipulator in the reference configuration.

The axis of rotation of the revolute joint at B makes an angle of 30° with the positive z axis of frame S , in a counter-clockwise direction as viewed in figure 1. This axis of rotation is indicated in this figure by the cyan coloured arrow through point B . The direction of the arrow indicates the direction of a positive rotation of the revolute joint, about the arrow line using a right hand rule.

Similarly, the axis of rotation of the revolute joint at D makes an angle of 45° with the positive z axis of frame S , in a counter-clockwise direction as viewed in figure 1, but this is true only when the manipulator is in the reference configuration. This axis of rotation is indicated in this figure by the cyan coloured arrow through point D . This arrow also shows the direction of a positive rotation of the revolute joint with the manipulator in the reference configuration. Note that, even though the revolute joint at D is not fixed to link 1, its orientation changes as the orientation of link 1 changes, because any motion in the prismatic joint between this revolute joint and point C does not change the orientation of the revolute joint.

The axis of displacement of the prismatic joint is fixed to the tip of link 1 at C and therefore changes direction as link 1 rotates. In the reference configuration, however, the axis of displacement of the

prismatic joint is parallel to the z axis of frame S . This axis of displacement is indicated in figure 1 by the cyan coloured arrow through point C . The direction of the arrow indicates the direction of a positive displacement of the prismatic joint, which is in the direction of the positive z axis of frame S when the manipulator is in the reference configuration.

The axis of rotation of the revolute joint at the tip of link 2, at the origin of frame T , is parallel to the positive z axis of frame S , but this is, like in the case of the revolute joint at D , true only when the manipulator is in the reference configuration. This axis is fixed to link 2 and therefore rotates with link 2. The direction of a positive rotation of this revolute joint is in the positive z axis of frame S using a right hand rule, with the system in the reference configuration. (No cyan arrow is shown in the figure for this revolute joint.)

To help to get a clearer picture of the operation of the prismatic joint, a second schematic drawing of the manipulator is shown in figure 2, with the only difference between this figure and figure 1 that the prismatic joint has been given a positive displacement. This of course means that the manipulator is no longer in the reference configuration in figure 2. This figure shows that, with such a displacement of the prismatic joint, the locations (but not the orientations) of the axes of rotation of the revolute joints at D and T have changed.

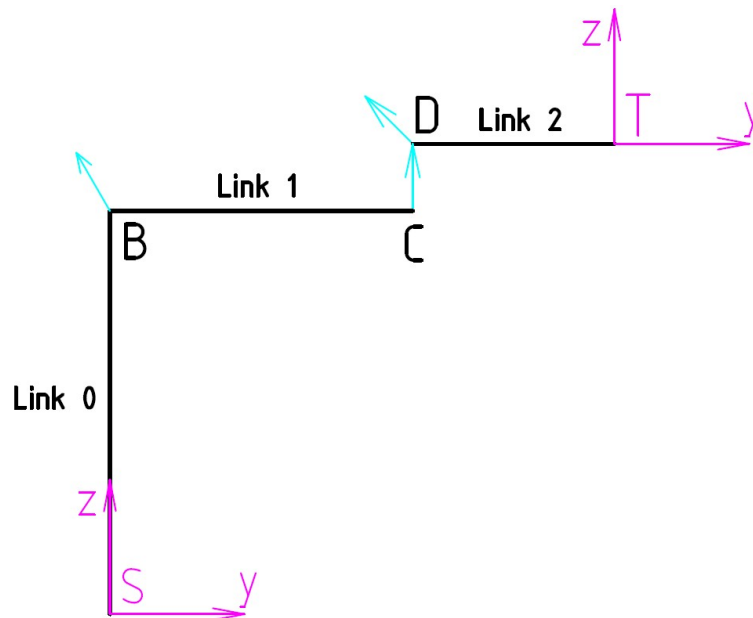


Figure 2: The manipulator in a configuration where the prismatic joint has moved through some positive displacement, with no rotation in the revolute joints at B and D .

The lengths of links 0, 1 and 2 are 0.6, 0.43 and 0.21 m, respectively.

The joint variables of the manipulator are $\theta_1, \theta_2, \theta_3, \theta_4$, the joint angle of the revolute joint at B , the displacement of the prismatic joint and the joint angles of the revolute joints at D and T , respectively. This defines $\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \mathbb{S}^1 \times \mathbb{R} \times \mathbb{S}^1 \times \mathbb{S}^1 = Q$.

Write a MATLAB script that calculates the transformation matrix $g_{st}(\theta)$ for any motion of the manipulator away from the reference configuration, **using the product of exponentials formula**. Then use this script to calculate the numerical values of the elements of $g_{st}(\theta)$ for the following values of the joint variables:

- $\theta_1 = 59$ degrees
- $\theta_2 = 0.55$ m
- $\theta_3 = -107$ degree
- $\theta_4 = 45$ degrees.

At the instant the new configuration $g_{st}(\theta)$ determined by the joint variables listed above, is reached, the joint velocities are

$$\dot{\theta} = \begin{bmatrix} 0.4 \\ -0.1 \\ 0.28 \\ 0.3 \end{bmatrix},$$

in rad/s or m/s.

Let q_T be the origin of the tool frame. Now expand your MATLAB script to do the following:

1. Calculate the instantaneous velocity vector $v_{q_T} \in \mathbb{R}^{3 \times 1}$ in spatial frame components.
2. Calculate the body velocity of the end effector frame $V_{st}^b \in \mathbb{R}^{6 \times 1}$.
3. With the manipulator in the current configuration, calculate the joint torques (six different answers) for the following six separate cases F_1 to F_6 of end effector wrench loading (in N and Nm), specified below in end effector body frame components. Ignore any possible effect of accelerations.
4. Calculate the corresponding six wrenches expressed in spatial frame components, to F_1 to F_6 , with the manipulator in the current configuration.
5. Check whether the calculation of the six sets of joint torques based on the six wrenches expressed in spatial frame components renders the same results as those calculated in part 3.

$$F_1 = \begin{bmatrix} 100 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad F_2 = \begin{bmatrix} 0 \\ 100 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad F_3 = \begin{bmatrix} 0 \\ 0 \\ 100 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad F_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 100 \\ 0 \\ 0 \end{bmatrix} \quad F_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 100 \\ 0 \end{bmatrix} \quad F_6 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 100 \end{bmatrix}$$

You need to report on this assignment in both a Moodle homework assignment and a Moodle homework feedback. The instructions for this will be made available on Moodle.